

Tab 1

RANDOM NUMBER GUESSING GAME

C Programming Project Report

Project Title: Random Number Guessing Game

Programming Language: C

Student Name: Prince

Course: B.Tech / AIML

Year: 1st Year

Academic Year: 2026–27

1. Introduction

The Random Number Guessing Game is a simple C programming project in which the computer generates a random number and the user tries to guess that number.

The program takes the user's guess as input and compares it with the randomly generated number. It then displays whether the guessed number is correct or not.

This project helps in understanding basic concepts of C programming such as variables, input/output, conditional statements, loops, and random number generation.

2. Objective

The main objectives of this project are:

- To understand the basic concepts of C programming.
 - To generate a random number using C.
 - To take input from the user.
 - To compare the user's guess with the generated number.
 - To use conditional statements and loops.
 - To develop a simple interactive console-based game.
-

3. Technology Used

Programming Language: C

Development Environment: Online C Compiler

Concepts Used:

- Variables
 - `printf()`
 - `scanf()`
 - `if-else`
 - Loops
 - Random number generation
 - Operators
-

4. Working of the Project

The program generates a random number within a particular range.

The user enters a number as a guess. The program compares the entered number with the randomly generated number.

If the guess is correct, the program displays a success message.

If the guess is incorrect, the program provides appropriate feedback and allows the user to continue guessing, depending on the program logic.

The game continues until the user guesses the correct number or the program reaches its defined condition.

5. Algorithm

1. Start the program.
 2. Generate a random number.
 3. Ask the user to enter a guess.
 4. Read the user's guess.
 5. Compare the guess with the random number.
 6. If both numbers are equal, display a success message.
 7. If they are not equal, display the appropriate message.
 8. Repeat the process if required.
 9. Stop the program.
-

6. Source Code

The complete source code of the project is provided in the `main.c` file in the GitHub repository.

File: `main.c`

```
#include <stdio.h>

#include <stdlib.h>

#include <time.h>

int main() {

    int randomNumber=(rand()%100)+1;

    int num_of_guessed =0;

    int guessed;

do{

    printf("guess the no");

    scanf("%d",&guessed);

    num_of_guessed ++;

    if(guessed > randomNumber){

        printf("guess lower no\n");

    }else{

        printf("guess higher no\n");

    }

}while(guessed!=randomNumber);

    printf("you guessed the no in %d guesses",num_of_guessed);
```

```
    return 0;  
}
```

7. Sample Output

- Example:

3:54 0.97 KB/S 64

Online C C...
programiz.com

Programiz
C Online Compiler

Programiz PRO

main.c Run

```
1 // Online C compiler (editor)
2 // Write and run C online using this editor.
3
4 #include <stdio.h>
5 #include <stdlib.h>
6 #include <time.h>
7 int main() {
8     int randomNumber=(rand()%100)+1;
9     int num_of_guessed =0;
10    int guessed;
11    do{
12        printf("guess the no");
13        scanf("%d",&guessed);
14        num_of_guessed ++;
15        if(guessed > randomNumber){
16            printf("guess lower no\n");
17        }else{
18            printf("guess higher no\n");
19        }
20    }while(guessed!=randomNumber);
21    printf("you guessed the no in %d
    guesses",num_of_guessed);
22
23
24
25    return 0;
26 }
```

Output

```
guess the no12
guess higher no
guess the no12
guess higher no
guess the no46
guess higher no
guess the no76
guess higher no
guess the no63
guess higher no
guess the no98
guess lower no
guess the no83
guess higher no
guess the no89
guess lower no
guess the no87
guess lower no
guess the no85
guess lower no
guess the no84
guess higher no
you guessed the no in 11 guesses
```

8. Advantages

- Easy to understand and use.
- Helps beginners learn C programming.

- Demonstrates random number generation.
 - Provides practical understanding of conditional statements and loops.
 - Can be extended with additional features.
-

9. Limitations

- It is a basic console-based project.
 - It does not have a graphical user interface.
 - The game has limited functionality compared with advanced games.
-

10. Future Scope

The project can be improved by adding:

- Difficulty levels.
 - Limited number of attempts.
 - Score calculation.
 - High-score system.
 - Hints such as "Too High" or "Too Low".
 - Multiple rounds.
 - Graphical user interface.
-

11. Conclusion

The Random Number Guessing Game is a simple and useful beginner-level C programming project. It demonstrates important programming concepts such as user input, conditional statements, loops, and random number generation.

Developing this project helped in understanding how basic C programming concepts can be combined to create an interactive application.

12. GitHub Repository

The complete project source code and project documentation are available in the GitHub repository.

Repository Files:

- [main.c](#) — Source code
- [README.md](#) — Project information
- [Random_Number_Guessing_Project_Report.pdf](#) — Project report

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    if(guessed > randomNumber){

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    }

}while(guessed!=randomNumber);

    printf("you guessed the no in %d guesses",num_of_guessed);
```

```
    return 0;  
}
```

7. Sample Output

- Example:

The screenshot shows the Programiz Online C Compiler interface. At the top, there's a status bar with the time 3:54, a WhatsApp icon, network speed 0.97 KB/S, signal strength, and battery level 64%. Below this is a header with a close button (X), a dropdown arrow, the title "Online C C... programiz.com", a share icon, a bookmark icon, and a menu icon (three dots). The main area has the Programiz logo and "C Online Compiler" text on the left, and a "Programiz PRO" button on the right. The central editor shows a C program named "main.c" with a "Run" button. The code is as follows:

```

1 // Online C compiler (editor)
2 // Write and run C online using this editor.
3
4 #include <stdio.h>
5 #include <stdlib.h>
6 #include <time.h>
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20    }while(guessed!=randomNumber);
21    printf("you guessed the no in %d
    guesses",num_of_guessed);
22
23
24
25    return 0;
26 }
  
```

The output on the right shows the program's execution: "guess the no12", "guess higher no", "guess the no12", "guess higher no", "guess the no46", "guess higher no", "guess the no76", "guess higher no", "guess the no63", "guess higher no", "guess the no98", "guess lower no", "guess the no83", "guess higher no", "guess the no89", "guess lower no", "guess the no87", "guess lower no", "guess the no85", "guess lower no", "guess the no84", "guess higher no", and finally "you guessed the no in 11 guesses".

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